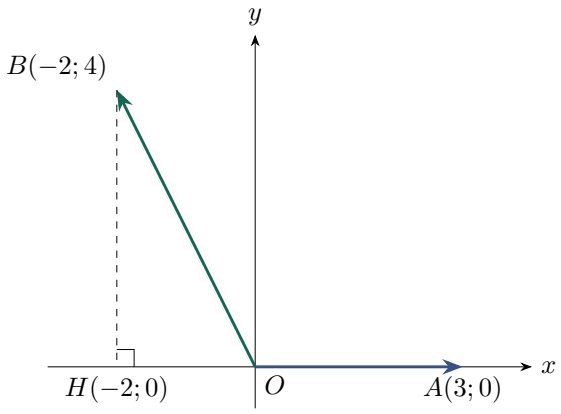
3.1 Projection and angle

Definition and results

For two non-zero vectors $\vec{u}, \vec{v}$, their scalar product is the real number $\vec{u} \cdot \vec{v} = \|\vec{u}\| \, \|\vec{v}\| \cos\theta$, where $\theta \in [0; \pi]$ is their non-oriented angle. With a zero vector, the product is 0. If H is the projection of B on (OA), $\overrightarrow{OA} \cdot \overrightarrow{OB}$ is $OA \times OH$ when H is on the ray [OA), and its opposite otherwise.

Status and justification. Definition; the equivalence with projection follows from the cosine in right-angled triangles, with a negative algebraic projection for an obtuse angle.

Worked example. For two vectors with norms 3 and 4 forming 60 °the product is $\dfrac{3 \times 4 \times 1}{2} = 6$. With 120 °, it equals -6. If $A(3; 0)$ and $B(-2; 4)$, the projection is $H(-2; 0)$ and the product is -6.



The projection H is on the negative side of the horizontal axis. Thus $\overrightarrow{OA} \cdot \overrightarrow{OB} = 3 \times (-2) = -6$.

Watch out

The result is a number, not a vector. A right angle corresponds to 0, not 1.

Immediate application. Two vectors have norms 2 and 5 and form 90 °, then 180 °. Calculate the products.

3.2 Coordinates, Norm and Orthogonality

Definition and results

In an orthonormal basis, for $\vec{u} = (x; y)$ and $\vec{v} = (x\prime; y\prime)$, $\vec{u} \cdot \vec{v} = xx\prime + yy\prime$ and $\|\vec{u}\|^2 = x^2 + y^2$. The vectors are orthogonal if and only if their scalar product is zero. For a basis $(\vec{i}, \vec{j})$ orthonormal, $x = \vec{u} \cdot \vec{i}$ and $y = \vec{u} \cdot \vec{j}$.

Status and justification. Properties proved using bilinearity: expand $(x\vec{i} + y\vec{j}) \cdot (x\prime\vec{i} + y\prime\vec{j})$ and use $\vec{i} \cdot \vec{i} = \vec{j} \cdot \vec{j} = 1$, $\vec{i} \cdot \vec{j} = 0$.

Worked example. $\vec{u} = (2; 3)$, $\vec{v} = (3; -2)$: product 6 − 6 = 0. Both vectors are orthogonal. Their norm is $\sqrt{13}$. The product of (2; 3) with (1; 0) is 2: we recover the x-coordinate.

Watch out

The usual formula assumes an orthonormal basis. Collinearity and orthogonality are different properties.

Immediate application. For which real t are the vectors $(t; 2)$ and $(3; -6)$ orthogonal?

3.3 Vector calculation and identities

Definition and results

The scalar product is symmetric and bilinear: $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$ and $(\lambda\vec{u}) \cdot \vec{v} = \lambda(\vec{u} \cdot \vec{v})$. Thus $\|\vec{u} \pm \vec{v}\|^2 = \|\vec{u}\|^2 \pm 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2$. We deduce $\vec{u} \cdot \vec{v} = \dfrac{\|\vec{u} + \vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2}{2}$.

Status and justification. Bilinearity accepted from the geometric definition; symmetry follows of the cosine. The identities are proved by expanding $(\vec{u} \pm \vec{v}) \cdot (\vec{u} \pm \vec{v})$.

Worked example. If $\|\vec{u}\| = 3$, $\|\vec{v}\| = 4$ and $\|\vec{u} + \vec{v}\| = 5$, the product is $\dfrac{25 - 9 - 16}{2} = 0$. We deduce orthogonality without coordinates.

Watch out

In general, $\|\vec{u} + \vec{v}\| \neq \|\vec{u}\| + \|\vec{v}\|$. The identities concern the squares of the norms.

Immediate application. We are given $\|\vec{u}\| = 2$, $\|\vec{v}\| = 3$, $\vec{u} \cdot \vec{v} = 1$. Calculate $\|\vec{u} - \vec{v}\|$.

3.4 Al-Kashi's theorem and choosing a method

Definition and results

In an ABC triangle, $BC^2 = AB^2 + AC^2 - 2AB\,AC\cos\widehat{BAC}$. This formula allows calculate the third side from two sides and the included angle, or determine the cosine of an angle from the three sides.

Status and justification. By Chasles, $\overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB}$. Expanding its scalar square gives $BC^2 = AC^2 + AB^2 - 2AB \cdot AC$; introduce the expression with the cosine.

Worked example. $AB = 5$, AC = 3 and $\widehat{BAC} = 60°$: $BC^2 = 25 + 9 - 15$ = 19, so $BC = \sqrt{19}$. With three sides, isolate the cosine before approximating the angle on a calculator, if allowed.

Watch out

The angle is the one between the two sides in the product term. The minus sign is essential.

Immediate application. A triangle has $AB = AC$ = 4 and $BC = 4\sqrt{2}$. Determine angle A.

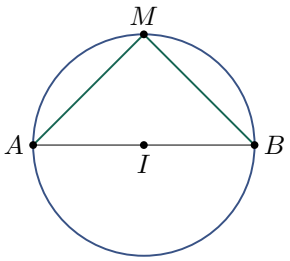
3.5 The circle with diameter AB

Definition and results

For two distinct points A and B with midpoint I, $\overrightarrow{MA} \cdot \overrightarrow{MB} = MI^2 - \dfrac{AB^2}{4}$. The set of points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB}$ = 0 is the circle with centre I and radius $\dfrac{AB}{2}$, points A and B included.

Status and justification. Write $\overrightarrow{MA} = \overrightarrow{MI} + \overrightarrow{IA}$ and $\overrightarrow{MB} = \overrightarrow{MI} - \overrightarrow{IA}$. The cross terms cancel ; there remain $MI^2 - IA^2$. Since $IA = \dfrac{AB}{2}$, setting it equal to zero is equivalent to the metric equation of the circle.

Worked example. $A(-2; 0)$ and $B(2; 0)$: I is the origin. The locus is $x^2 + y^2$ = 4. For $M(0; 1)$, the product is 1 $- 4 = -3$; for $M(0; 2)$, it equals 0.



Circle with diameter AB: the vectors $\overrightarrow{MA}$ and $\overrightarrow{MB}$ are orthogonal.

Watch out

The endpoints belong to the locus even though "right-angled triangle" does not apply to a degenerate triangle.

Immediate application. For $A(0;0)$, $B(6;0)$, find the points M such that $\overrightarrow{MA} \cdot \overrightarrow{MB} = 7$.